

MATERIAL SAFETY DATA SHEET

SECTION 1 CHEMICAL IDENTIFICATION

PRODUCT NAME: Mancozeb + Cymoxanil (64%+8%) WP

COMPANY: Ningbo Greenidea Chemicals Co., Ltd.

CHEMICAL NAME: Mancozeb: [manganese ethylenebis(dithiocarbamate) (polymeric) complex with zinc salt]
Cymoxanil: [1-(2-cyano-2-methoxyiminoacetyl)-3-ethylurea]

SYNONYMS: None

CAS RN: Mancozeb: 8018-01-7
Cymoxanil: 57966-95-7

CHEMICAL CLASS: Mancozeb: Alkylenebis (dithiocarbamate)
Cymoxanil: Acetimide

USE: Agricultural fungicide

FORMULATION: Wettable Powder (WP)

SECTION 2 COMPOSITION/INFORMATION ON INGREDIENTS

Ingredient	CAS No.	Proportion (% w/w)
Mancozeb	8018-01-7	64%
Cymoxanil	57966-95-7	8%
Other inert ingredients	(non hazardous)	28%

SECTION 3 HAZARD INFORMATION

Irritating to eyes and respiratory system.

May cause sensitization by inhalation and skin contact.

Very toxic to aquatic organisms, may cause long-term adverse effects in the aquatic environment.

SECTION 4 FIRST AID MEASURES

General advice: Never give anything by mouth to an unconscious person. If breathing is irregular or stopped, administer artificial respiration.

Inhalation: Move to fresh air. Consult a physician.

Skin contact: Wash off immediately with soap and plenty of water. If skin irritation persists, call a physician.

Eye contact: Rinse thoroughly with plenty of water, also under the eyelids. If eye irritation persists, consult a specialist.

Ingestion: Drink 1 or 2 glasses of water. Do not induce vomiting without medical advice. Consult a physician.

SECTION 5 FIRE FIGHTING MEASURES

Suitable extinguishing media:

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Extinguishing media which must not be used for safety reasons:

High volume water jet, (contamination risk).

Special protective equipment for fire-fighters:

In the event of fire, wear self-contained breathing apparatus.

Further information:

If area is heavily exposed to fire and if conditions permit, let fire burn itself out since water may increase the area contaminated. Cool containers/tanks with spray water.

SECTION 6 ACCIDENTAL SPIL/RELEASE MEASURES

Personal precautions: Wear personal protective equipment.

Environmental precautions:

Do not flush into surface water or sanitary sewer system.

Methods for cleaning up:

Clean up promptly by scoop or vacuum. Use approved industrial vacuum cleaner for removal. Shovel into suitable container for disposal.

SECTION 7 HANDLING AND STORAGE

Handling

Advice on safe handling: Provide appropriate exhaust ventilation at places where dust is formed. Wear personal protective equipment.

Advice on protection against fire and explosion: Keep away from heat and sources of ignition.

Storage

Requirements for storage areas and containers: Keep containers tightly closed in a dry, cool and well-ventilated place. Store in a place accessible by authorized persons only. Keep away from food, drink and animal feedingstuffs.

Storage temperature: < 49 °C

SECTION 8 EXPOSURE CONTROL/PERSONAL PROTECTION

Engineering measures

Ensure adequate ventilation, especially in confined areas.

Personal protective equipment

Respiratory protection: Half mask with a particle filter P2 (EN143)

Hand protection: Material: nitrile rubber

Wearing time: 480 min

Do not wear cotton or leather gloves. Take note of the information given by the producer concerning permeability and break through times, and of special workplace conditions (mechanical strain, duration of contact).

Eye protection: safety glasses

Skin and body protection: lightweight protective clothing

Hygiene measures: Wash hands and face before breaks and immediately after handling the product. When using, do not eat, drink or smoke. Do not breathe dust. Avoid contact with the skin and the eyes.

Protective measures: All chemical protective clothing should be visually inspected prior to use. Clothing and gloves should be replaced in case of chemical or physical damage or if contaminated.

SECTION 9 PHYSICAL AND CHEMICAL PROPERTIES

Form	Powdered solid
Color	Off white
Odor	None
pH	5.0 – 9.0
Viscosity	Not Applicable
Bulk density	600 kg/m ³

Solubility in Water	Dispersible
Flash point	Not applicable
Explosive properties:	Not explosive

SECTION 10 STABILITY AND REACTIVITY

Materials to avoid: acids

Hazardous decomposition products:

Sulphur compounds, nitrogen oxides (NO_x), Carbon oxides

Hazardous reactions:

No decomposition if stored and applied as directed.

SECTION 11 TOXICOLOGICAL INFORMATION

Acute oral toxicity:	LD50/rat: > 2000 mg/kg.
Acute inhalation toxicity:	LC50/4 h/rat: > 5.4 mg/L.
Acute dermal toxicity:	LD50/rat: > 2000 mg/kg.
Skin irritation:	No skin irritation.
Eye irritation:	Slight eye irritation.
Sensitization:	Not a skin sensitizer.

SECTION 12 ECOLOGICAL INFORMATION

Mancozeb Technical:

Birds: In ten-day dietary studies, there were no mortalities in mallard ducks at 6400 mg/kg daily, and in Japanese quail at 3200 mg/kg daily.

Fish: LC50 (48 h) for goldfish 9.0, rainbow trout 2.2, carfish 5.2, carp 4.0 mg/L.

Bees: LC50 0.193 mg/bee.

Cymoxanil technical:

Bird: Acute oral LD50 for mallard duck and bobwhite quail > 2250 mg/kg. 8-day dietary LC50 for mallard ducks and bobwhite quail > 5620 mg/kg.

Fish: Acute LC50 (96 h) for rainbow trout 61, bluegill fish 29, common carp 91 mg/l.

Daphnia: Acute LC50 (48 h) for Daphnia magna 27.0 mg/l.

Bees: Acute LD50 (contact, 48 h) > 25 µg/bee, LC50 (oral, 48 h) > 1000 ppm.

Earthworm: LD50 (14 d) > 2208 mg/kg soil.

SECTION 13 DISPOSAL INFORMATION
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Product disposal:

In accordance with local and national regulations. Must be incinerated in a suitable incineration plant holding a permit delivered by the competent authorities. Do not contaminate ponds, waterways or ditches with chemical or used container.

Contaminated packaging disposal:

Do not re-use empty containers.

SECTION 14 TRANSPORTATION INFORMATION
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ADR

Class: 9

Packaging group: III

Classification Code: M7

HI No.: 90
UN-No: 3077
Labelling No.: 9
Proper shipping name: Environmentally hazardous substance, solid, n.o.s.
(contains Mancozeb)

IATA_C

Class: 9
Packaging group: III
UN-No: 3077
Labelling No.: 9MI
Proper shipping name: Environmentally hazardous substance, solid, n.o.s.
(contains Mancozeb)

SECTION 15 REGULATORY INFORMATION
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Labelling according to EEC Directive

Symbol(s):	Xn	Harmful
	N	Dangerous for the environment

Hazardous components: Mancozeb
Cymoxanil

R-phrase(s):	R36/37	Irritating to eyes and respiratory system.
	R42/43	May cause sensitization by inhalation and skin contact.
	R50/53	Very toxic to aquatic organisms, may cause long-term adverse effects in the aquatic environment.

S-phrase(s):	S22	Do not breathe dust.
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S26	In case of contact with eyes, rinse immediately with plenty of water and seek medical advice.
S36/37/39	Wear suitable protective clothing, gloves and eye/face protection.
S45	In case of accident or if you feel unwell, seek medical advice immediately (show label where possible).
S63	In case of accident by inhalation: remove casualty to fresh air and keep at rest.
S35	This material and its container must be disposed of in a safe way.
S57	Use appropriate containment to avoid environmental contamination.

SECTION 16 OTHER INFORMATION

Revise Date:

October 22, 2015

This MSDS summarises our best knowledge of the health and safety hazard information of the product and how to safely handle and use the product in the workplace. Each user should read this MSDS and consider the information in the context of how the product will be handled and used in the workplace including in conjunction with other products.